

WASP-177b

R-2344

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-177_210805 · created 2026-08-19T00:37:45+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459431.9494 +/- 0.0073 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.048
Transit depth (Rp/Rs)^2	2.86 +/- 1.61 [%]
Orbital Inclination (inc)	87.57 +/- 2.59
Airmass coefficient 1 (a1)	0.886 +/- 0.017
Airmass coefficient 2 (a2)	0.097 +/- 0.032
Scatter in the residuals of the lightcurve fit is	1.93 %
Best Comparison Star	#1 - [146, 426]
Optimal Aperture	4.48
Optimal Annulus	12.17
Transit Duration (day)	0.091 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.169 ± 0.048 vs 0.1618 (deviation 0.15σ)
Duration fit vs published	0.091 d vs 0.0721 d (deviation 0.86σ)
Mid-time O–C	17.7 min
Depth SNR	3.5
Residual scatter	1.93 %

Light curve

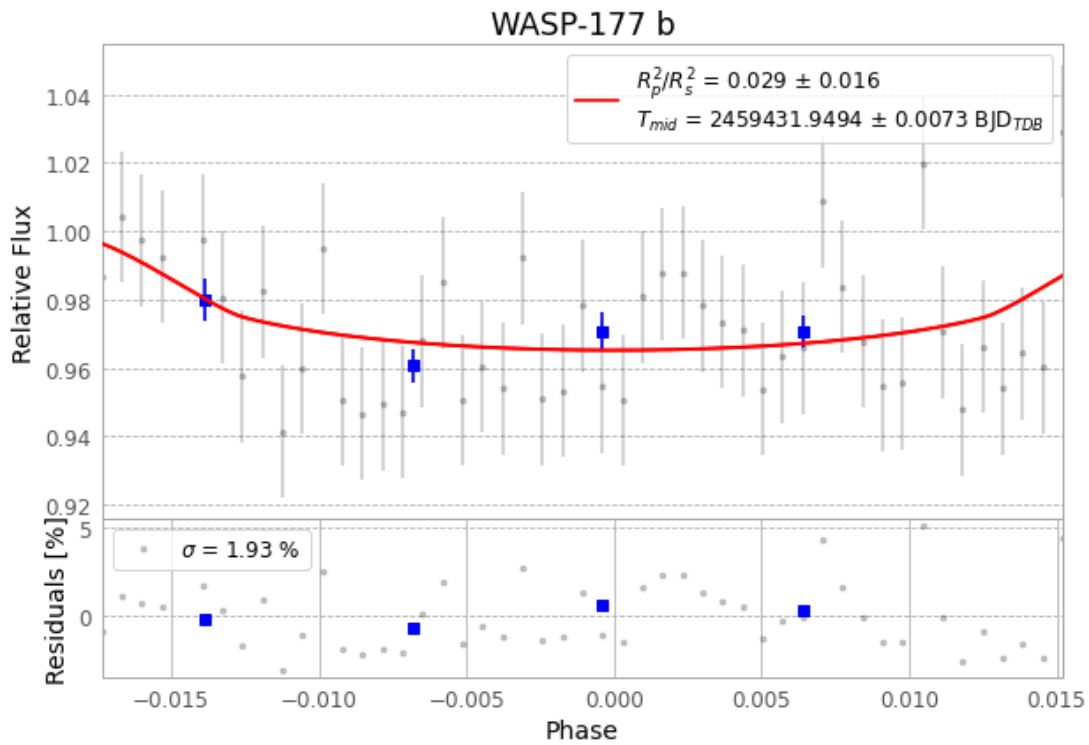


Figure 1 — FinalLightCurve_WASP-177 b_2021-08-05.png (SHA-256 5254a695efea08cd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [354, 101], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 261dc04f59c0fb0f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

49 FITS frames (32 MB), WASP-177210805092112.FITS → WASP-177210805114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-177-210805.log (8bdcdbde6270...), FinalLightCurve_WASP-177 b_2021-08-05.png (5254a695efea...), FinalLightCurve_WASP-177 b_2021-08-05.csv (0ed4d5166b24...)

Compute resources

Wall time 97.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 00:37Z.